

UNIT1: Part 2

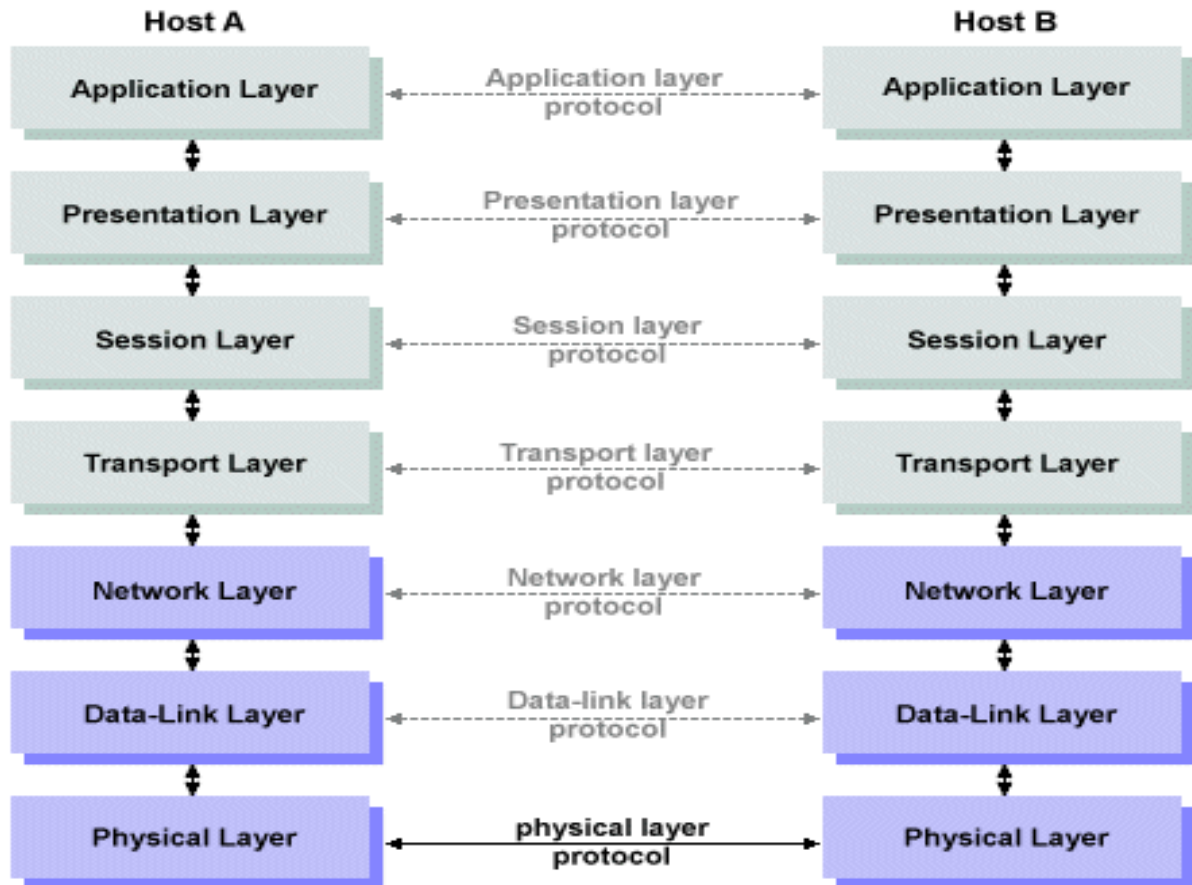
OSI and TCP/IP models, Networking Devices

OSI MODEL

- The OSI (Open Systems Interconnection) Model is a **set of rules that explains how different computer systems communicate over a network.**
- OSI Model was developed by the **International Organization for Standardization (ISO).**
- The **OSI Model consists of 7 layers** and each layer has specific functions and responsibilities.
- This layered approach makes it easier for different devices and technologies to work together.

Why OSI Model is Needed

- Breaks complex network communication into **simple layers**
- Helps in **standardization** of networking technologies
- Makes **troubleshooting easier** (identify problem layer-wise)
- Enables **interoperability** between different vendors



Brief Overview of 7 Layers

Layer No.	Layer Name	Key Purpose
7	Application	User interaction
6	Presentation	Data format & security
5	Session	Session management
4	Transport	End-to-end delivery
3	Network	Routing & logical addressing
2	Data Link	Physical addressing
1	Physical	Bit transmission

Layer 1: Physical Layer

- The lowest layer of the OSI reference model is the Physical Layer.
- It is responsible for the **actual physical connection between the devices**.
- The physical layer contains information in the **form of bits**.

How it works?

- Physical Layer is responsible for **transmitting individual bits from one node to the next**.
- When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together.
- **Physical layer devices : Hub, Repeater, Modem, and Cables.**

Functions of the Physical Layer

- **Bit Synchronization:** The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at the bit level.
- **Bit Rate Control:** The Physical layer also defines the transmission rate i.e. the number of bits sent per second.
- **Physical Topologies:** Physical layer specifies how the different, devices/nodes are arranged in a network i.e. bus topology, star topology, or mesh topology.
- **Transmission Mode:** Physical layer also defines how the data flows between the two connected devices. The various transmission modes possible are Simplex, half-duplex and full duplex.

Layer 2: Data Link Layer (DLL)

- The data link layer is responsible for the **node-to-node delivery of the message**.
- The main function of this layer is to make sure data transfer **is error-free from one node to another**, over the physical layer.
- When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its MAC address.
- **Packet in the Data Link layer is referred to as Frame.**
- **Devices in DLL Layer: Switches and Bridges**
- The packet received from the Network layer is further divided into frames depending on the frame size of the NIC (Network Interface Card).
- DLL also **encapsulates Sender and Receiver's MAC address** in the header.
- The Receiver's MAC address is obtained by placing an ARP (Address Resolution Protocol) request onto the wire asking, "Who has that IP address?" and the destination host will reply with its MAC address.

Sublayers of Data Link Layer

- Logical Link Control (LLC)
- Media Access Control (MAC)

1. LLC (Logical Link Control) – Upper Sub-layer

- LLC provides **logical control of the data link** and acts as an **interface between the Network Layer and MAC layer**.

Functions of LLC

- **Protocol identification**
 - Identifies which Network Layer protocol is being used (IP, IPX, etc.)
- **Flow control**
 - Controls data flow between sender and receiver
- **Error control**

- Detects and may request retransmission of lost frames
- **Multiplexing**
Allows multiple network protocols to use the same MAC layer

Why LLC is Used?

- Makes Data Link Layer **independent of physical media**
- Provides a **standard interface** to the Network Layer
- Helps multiple protocols share the same network

Example

When a frame reaches Layer 2, LLC checks:

“Is this data for IPv4, IPv6, or some other protocol?”

2. MAC (Media Access Control) – *Lower Sub-layer*

MAC controls **how devices access the shared transmission medium** and handles **physical addressing**.

Functions of MAC

- **MAC addressing** (Source & Destination MAC)
- **Frame encapsulation**
- **Media access control**
 - Decides *who can transmit and when*
- **Error detection** using CRC
- **Collision handling** (e.g., CSMA/CD in Ethernet)

Why MAC is Used?

- Ensures **orderly transmission** on shared media
- Provides **unique identification** of devices
- Prevents or manages **data collisions**

Example

MAC decides: “Can I send data now, or is the channel busy?”

Functions of the Data Link Layer

- **Framing:**
 - Framing is a function of the data link layer.
 - It provides a way for a sender to transmit a set of bits that are meaningful to the receiver.
 - **This can be accomplished by attaching special bit patterns to the beginning and end of the frame.**
- **Physical Addressing:**
 - After creating frames, the Data link layer adds physical addresses (MAC addresses) of the sender and/or receiver in the header of each frame.
- **Error Control:**
 - The data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.
- **Flow Control:**
 - The data rate must be constant on both sides else the data may get corrupted thus, flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- **Access Control:**
 - When a single communication channel is shared by multiple devices, the MAC sub-layer of the data link layer helps to determine which device has control over the channel at a given time.

Layer 3: Network Layer

- The Network Layer works for the **transmission of data from one host to the other located in different networks.**
- It also takes care of **packet routing** i.e. selection of the shortest path to transmit the packet, from the number of routes available.
- The **sender and receiver's IP address are placed in the header** by the network layer.
- **Segment in the Network layer is referred to as Packet.**
- **Network layer devices: Routers and Switches**

Functions of the Network Layer

- **Routing:**
 - The network layer protocols determine which route is suitable from source to destination.
 - This function of the network layer is known as routing.
- **Logical Addressing:**
 - To identify each device inter-network uniquely, the network layer defines an addressing scheme.
 - The sender and receiver's IP addresses are placed in the header by the network layer.
 - Such an address distinguishes each device uniquely and universally.

Layer 4: Transport Layer

- The Transport Layer provides services to the application layer and takes services from the network layer.
- **The data in the transport layer is referred to as Segments.**
- It is responsible for the **end-to-end delivery of the complete message.**
- The transport layer also **provides the acknowledgment of the successful data transmission and re-transmits the data if an error is found.**
- **Protocols used in Transport Layer are TCP, UDP NetBIOS, PPTP.**
- At the sender's side, the transport layer receives the formatted data from the upper layers, performs Segmentation, and also implements Flow and error control to ensure proper data transmission.
- It also **adds Source and Destination port number in its header** and forwards the segmented data to the Network Layer.
- Generally, this destination port number is configured, either by default or manually.
- **Example:** when a web application requests a web server, it typically uses port number 80, because this is the default port assigned to web applications. Many applications have default ports assigned.
- At the Receiver's side, Transport Layer reads the port number from its header and forwards the Data which it has received to the respective application. **It also performs sequencing and reassembling of the segmented data.**

Functions of the Transport Layer

- **Segmentation and Reassembly:**
 - This layer accepts the message from the (session) layer and breaks the message into smaller units.
 - Each of the segments produced has a header associated with it.
 - The transport layer at the destination station reassembles the message.

- **Service Point Addressing:**

- To deliver the message to the correct process, the transport layer header includes a type of address called service point address or port address.
- Thus, by specifying this address, the transport layer makes sure that the message is delivered to the correct process.

Services Provided by Transport Layer

- Connection-Oriented Service
- Connectionless Service

Layer 5: Session Layer

- Session Layer in the OSI Model is responsible for the **establishment of connections, management of connections, terminations of sessions between two devices.**
- It also provides **authentication and security.**
- **Protocols** used in the Session Layer are **NetBIOS, PPTP.**

Functions of the Session Layer

- **Session Establishment, Maintenance, and Termination:**
 - The layer allows the two processes to establish, use, and terminate a connection.
- **Synchronization:**
 - This layer allows a process to **add checkpoints that are considered synchronization points in the data.**
 - These synchronization points help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely, and data loss is avoided.
- **Dialog Controller:**
 - The session layer allows two systems to start communication with each other in half-duplex or full duplex.

Layer 6: Presentation Layer

- The Presentation Layer is also called the **Translation layer**.
- The data from the application layer is extracted here and manipulated as per the required format to transmit over the network.
- **Protocols used in the Presentation Layer: TLS/SSL (Transport Layer Security / Secure Sockets Layer).**
- JPEG, MPEG, and GIF are standards or formats used for encoding data, which is part of the presentation layer's role.

Functions of the Presentation Layer

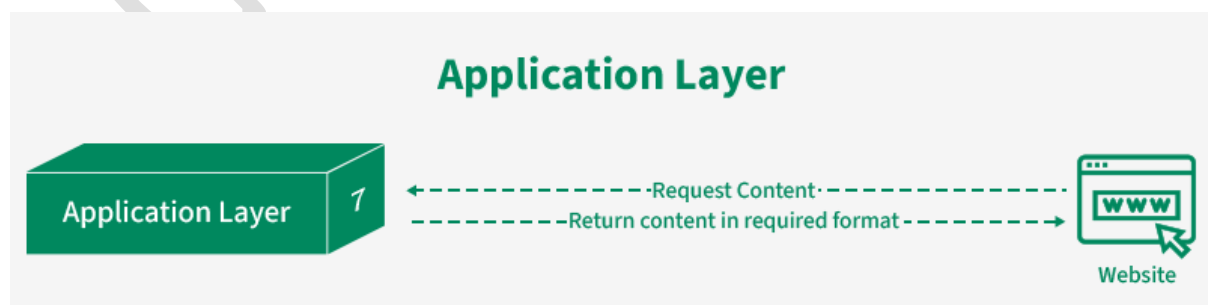
- **Translation:** For example, ASCII to EBCDIC.
- **Encryption/ Decryption:** Data encryption translates the data into another form or code. The encrypted data is known as the ciphertext, and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.
- **Compression:** Reduces the number of bits that need to be transmitted on the network.

Layer 7: Application Layer

At the very top of the OSI Reference Model stack of layers, we find the Application Layer which is implemented by the network applications.

These applications produce the data to be transferred over the network.

- This layer also serves as a window for the application services to access the network and for displaying the received information to the user.
- **Protocols used in the Application layer are SMTP, FTP, DNS, etc.**



Application Layer

Functions of the Application Layer

- **Network Virtual Terminal (NVT):** It allows a user to log on to a remote host.
- **File Transfer Access and Management (FTAM):** This application allows a user to access files in a remote host, retrieve files in a remote host, and manage or control files from a remote computer.
- **Mail Services:** Provide email service.
- **Directory Services:** This application provides distributed database sources and access for global information about various objects and services.

How Data Flows in the OSI Model?

- When we transfer information from one device to another, it travels through 7 layers of OSI model.
- First data travels down through 7 layers from the sender's end and then climbs back 7 layers on the receiver's end.

Data flows through the OSI model in a step-by-step process:

- **Application Layer:** Applications create the data.
- **Presentation Layer:** Data is formatted and encrypted.
- **Session Layer:** Connections are established and managed.
- **Transport Layer:** Data is broken into segments for reliable delivery.
- **Network Layer:** Segments are packaged into packets and routed.
- **Data Link Layer:** Packets are framed and sent to the next device.
- **Physical Layer:** Frames are converted into bits and transmitted physically.

Each layer adds specific information to ensure the data reaches its destination correctly, and these steps are reversed upon arrival.

EXAMPLE

We can understand how data flows through OSI Model with the help of an example mentioned below. Let us suppose, Person A sends an e-mail to his friend Person B.

- **Step 1: Person A** interacts with e-mail application like Gmail, outlook, etc. Writes his email to send. (This happens at Application Layer).
- **Step 2: At Presentation Layer**, Mail application prepares for data transmission like encrypting data and formatting it for transmission.
- **Step 3: At Session Layer**, there is a connection established between the sender and receiver on the internet.
- **Step 4: At Transport Layer**, Email data is broken into smaller segments. It adds sequence number and error-checking information to maintain the reliability of the information.
- **Step 5: At Network Layer**, addressing of packets is done in order to find the best route for transfer.
- **Step 6: At Data Link Layer**, data packets are encapsulated into frames, then MAC address is added for local devices and then it checks for error using error detection.
- **Step 7: At Physical Layer**, Frames are transmitted in the form of electrical/ optical signals over a physical network medium like ethernet cable or WiFi.

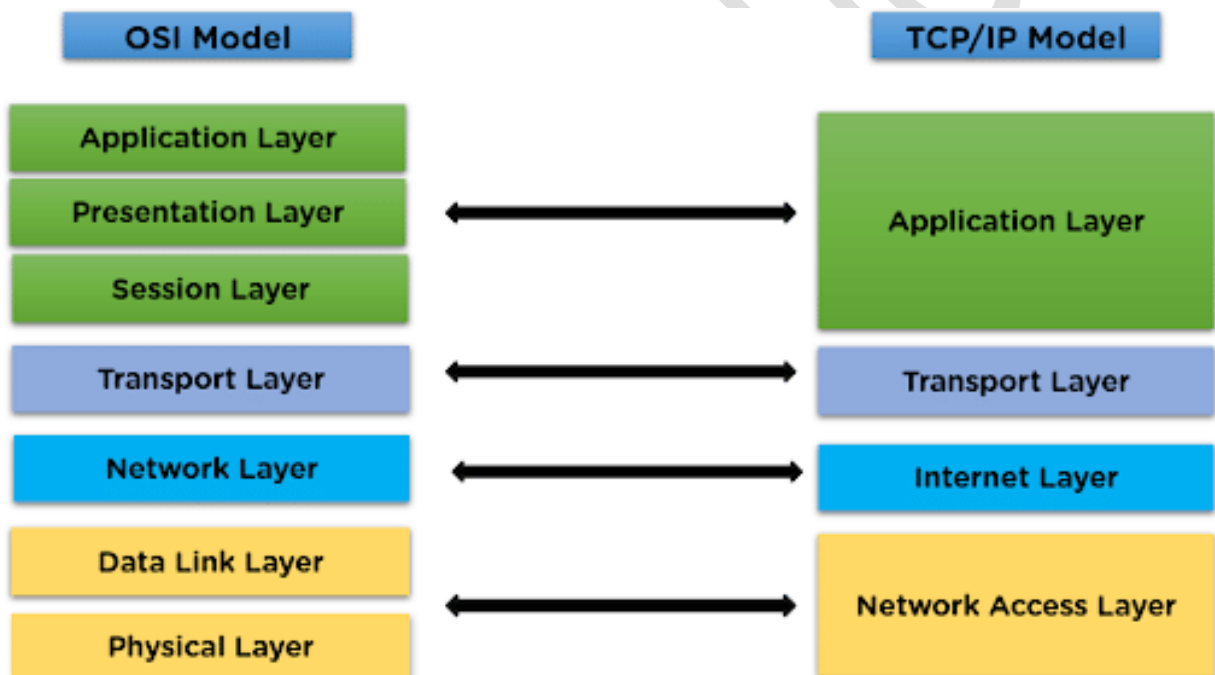
After the email reaches the receiver i.e. Person B, the process will reverse and decrypt the e-mail content. At last, the email will be shown on Person B email client.

TCP/IP Model

(Transmission Control Protocol / Internet Protocol)

- The **TCP/IP model** is a **layered network architecture** developed by **DARPA (USA)** for reliable communication over interconnected networks.
- It is the **foundation of the Internet** and defines how data is transmitted from a **source host** to a **destination host**.
- Unlike the OSI model (theoretical), the TCP/IP model is **practical and implementation-based**.

Layers of TCP/IP Model (4 Layers)



1. Application Layer

Purpose

- Provides **network services** directly to **end-user applications**
- Handles **data representation, session management, and communication**
- Combines **Application, Presentation, and Session** layers of the OSI model

Main Functions

- Data formatting and encoding
- User authentication
- Establishing and terminating communication sessions
- File transfer and email services

Protocols Used

- **HTTP / HTTPS** – Web browsing
- **FTP** – File transfer
- **SMTP** – Sending emails
- **POP3 / IMAP** – Receiving emails
- **DNS** – Domain name to IP address translation
- **Telnet / SSH** – Remote login

Example

When you type **www.google.com** in a browser:

- HTTP requests the webpage
- DNS converts the domain name into an IP address

2. Transport Layer

Purpose

- Provides **end-to-end communication** between sender and receiver
- Ensures data is delivered **correctly and in sequence**

Main Functions

- Segmentation and reassembly of data
- Flow control
- Error detection and recovery
- Port addressing (process-to-process delivery)

Protocols Used

TCP (Transmission Control Protocol): TCP is used when data must be error-free, complete, and in order. Examples: loading websites, downloading files, sending emails.

- Connection-oriented
- Reliable and ordered delivery
- Error checking and retransmission
- Used where accuracy is important

Example: File transfer, email, web pages

UDP (User Datagram Protocol): UDP is used when speed is more important than perfect accuracy. Examples: live streaming, online gaming, VoIP calls. Its features are:

- Connectionless
- Faster but unreliable
- No retransmission or ordering

Example: Video streaming, online games, VoIP

3. Internet Layer

Purpose

- Responsible for **logical addressing and routing**
- Transfers packets from source network to destination network

Main Functions

- Packet forwarding
- Logical (IP) addressing
- Routing between networks
- Fragmentation and reassembly of packets

Protocols Used

- **IP (IPv4 / IPv6)** – Logical addressing
- **ICMP** – Error reporting and diagnostics
- **ARP** – IP address to MAC address mapping
- **RARP** – MAC address to IP address mapping

Example

- Router uses IP addresses to decide the best path for data
- ping command uses ICMP

4. Network Access Layer

Purpose

- Responsible for the **physical transmission of data**
- Interfaces with the actual network hardware

Main Functions

- Framing
- Physical addressing (MAC address)
- Error detection at frame level
- Media access control

Technologies Used

- Ethernet
- Wi-Fi
- LAN, WAN
- MAC addresses
- Network Interface Cards (NIC)

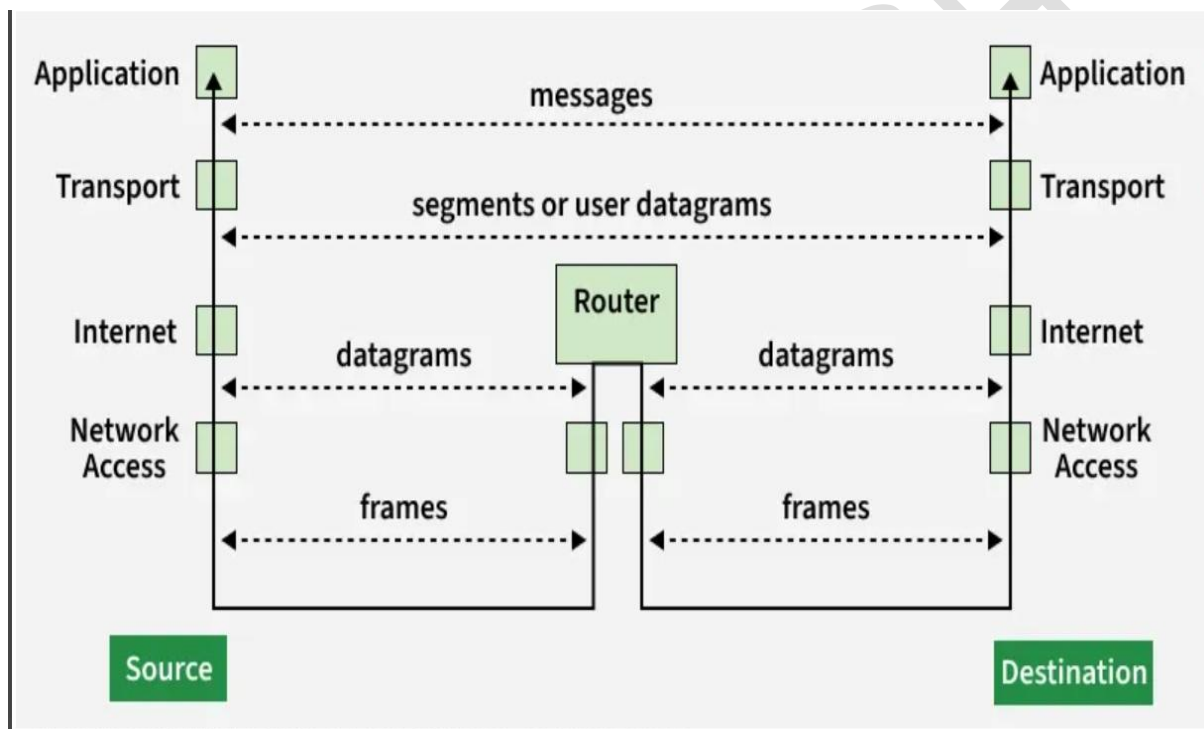
Example

Sending data over Ethernet cable or Wi-Fi router

Data Flow in TCP/IP Model

1. Application Layer creates data
2. Transport Layer divides data into segments
3. Internet Layer adds IP address and routes packets
4. Network Access Layer sends bits over physical medium

At receiver side, the process is **reversed**.



Comparison: TCP/IP vs OSI Model

Feature	TCP/IP Model	OSI Model
Number of layers	4	7
Nature	Practical	Theoretical
Developed by	DARPA	ISO
Usage	Internet	Education
Flexibility	High	Low

Advantages of TCP/IP Model

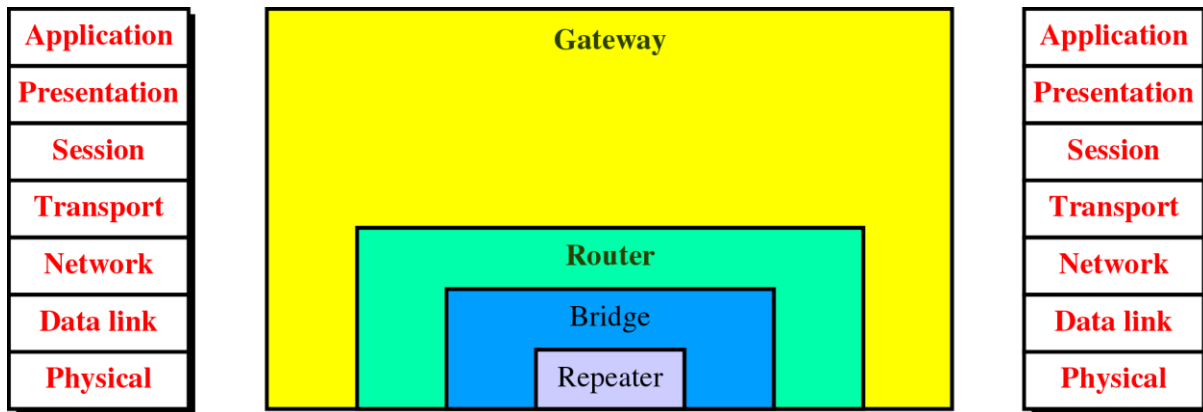
- Standard protocol of the Internet
- Highly scalable
- Platform independent
- Supports both reliable (TCP) and fast (UDP) communication

Disadvantages

- No clear separation of services, interfaces, and protocols
- Session and Presentation layers not defined separately
- Less secure by default

NETWORKING DEVICES

- Network devices are hardware components that help computers and other devices communicate over a network by forwarding, regenerating, or routing data.
- They operate at different layers of the OSI model depending on their function



Layer 1 – Physical Layer Devices

1) Hub

A hub is a basic networking device that **connects multiple computers in a LAN and sends data to all connected devices.**

- **Function:**
Provides a common connection point for network devices.

How it Works:

Receives data from one port and **broadcasts it to all ports** without checking the destination.

Key Points:

- Works at the Physical layer
- No intelligence
- Causes collisions
- Obsolete device

Example:

In an old computer lab, 10 computers are connected using a hub. When one computer sends data, all computers receive it.

Real-life Analogy:

Teacher announcing something loudly in a classroom.

2) Repeater

A repeater is a networking device used to regenerate and strengthen weak signals in a network.

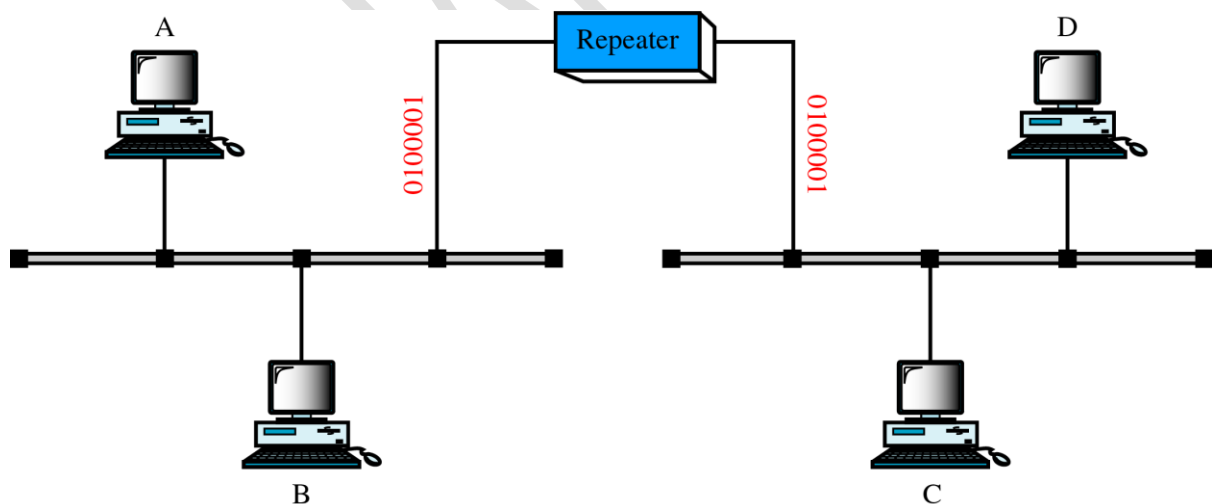
- ▶ A repeater installed on a link receives the signal before it becomes too weak or corrupted, regenerates the original pattern, and puts the refreshed copy back on the link.

Function:

Extends the physical length of a network.

How it Works:

Receives weak signals, **regenerates and retransmits** them to their original strength.

**Key Points:**

- Physical layer device
- Solves signal attenuation
- No data filtering

Example:

A repeater is placed between two distant buildings to boost network signals.

Real-life Analogy:

A loudspeaker that amplifies sound for distant listeners.

3) Modem (Modulator–Demodulator)

A modem is a device that converts digital data into analog signals and vice versa for Internet communication.

Function:

Connects a computer or network to an Internet Service Provider (ISP).

How it Works:

- **Modulation:** Digital → Analog
- **Demodulation:** Analog → Digital

Key Points:

- Required for Internet access
- Works at Physical layer

Example:

A home DSL modem connects your laptop to the Internet.

Real-life Analogy:

A language translator converting one language to another.

Layer 2 – Data Link Layer Devices

4) Bridge

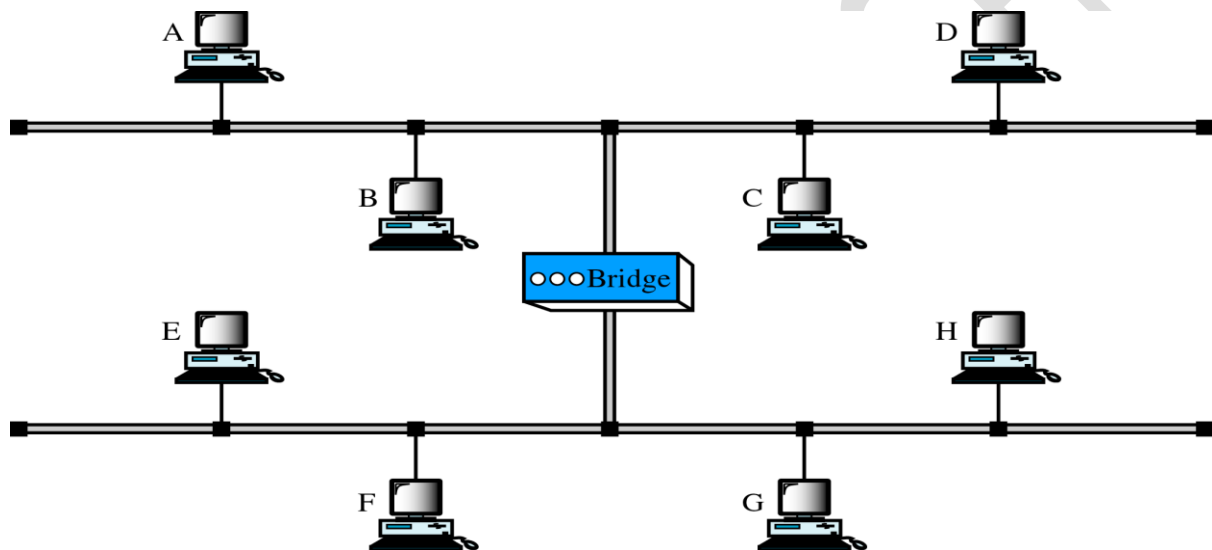
A bridge is a device that **connects two LAN segments and filters traffic using MAC addresses**.

- **Function:**

Reduces network traffic by dividing a LAN into segments.

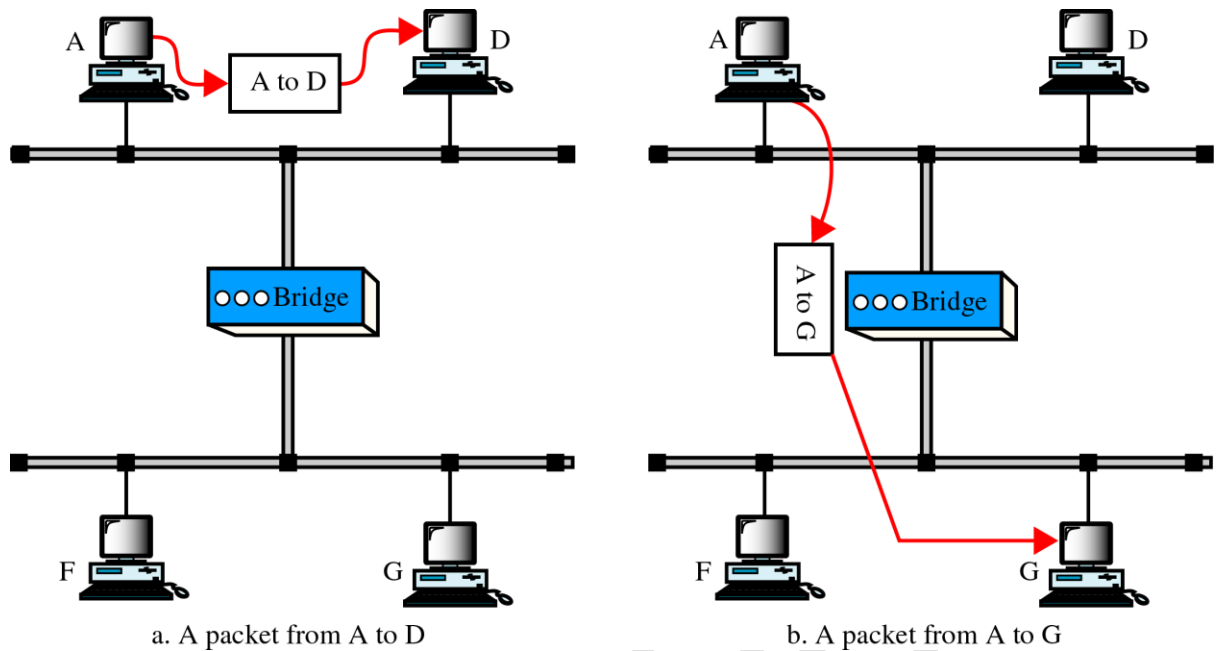
How it Works:

Checks **MAC addresses** and forwards frames only when necessary.



Key Points:

- Works at Data Link layer
- Reduces collisions
- Largely replaced by switches



Example:

Two departments in an office have separate LAN segments connected using a bridge.

Real-life Analogy:

A security guard controlling movement between two buildings.

5) Switch

A switch is an intelligent networking device that **connects multiple devices within a LAN**.

Function:

Efficiently forwards data to the correct destination device.

How it Works:

Uses a **MAC address table** to send data only to the intended port.

Key Points:

- Data Link layer device
- Faster and more secure than hub

- Reduces collisions

Example:

In a modern office, 20 computers are connected through a switch.

Real-life Analogy:

A postman delivering letters to specific houses.

Layer 3 – Network Layer Device

6) Router

A router is a networking device that **connects different networks and routes data packets between them.**

- **Function:**
Selects the best path for data transmission.

How it Works:

- Uses **IP addresses and routing tables** to forward packets.

Key Points:

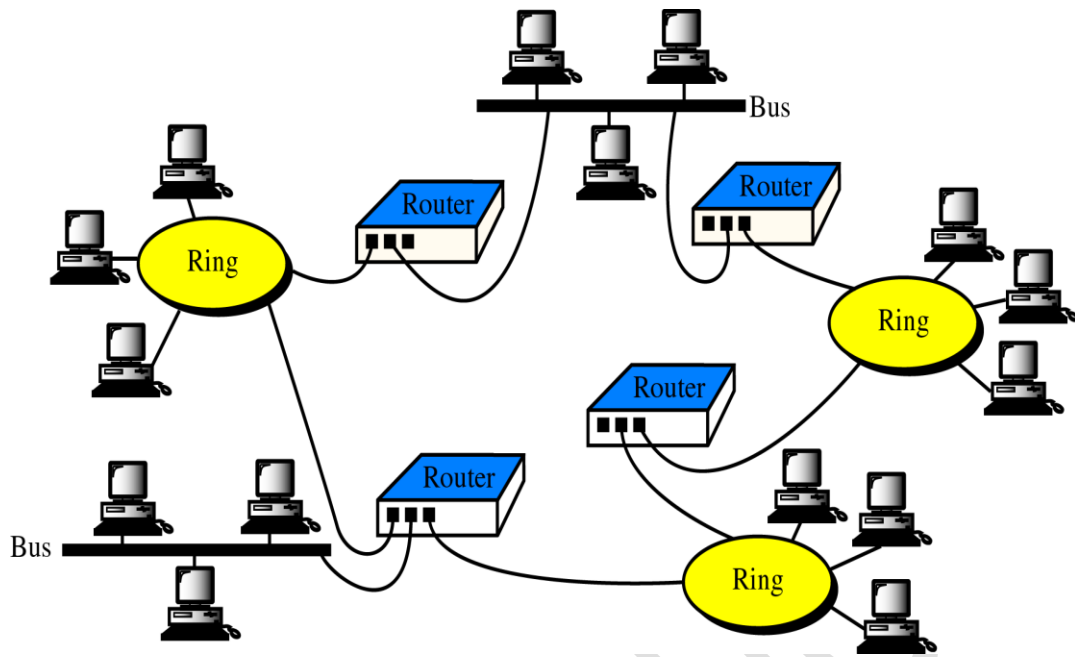
- Works at Network layer
- Connects LAN to Internet
- Prevents broadcast storms

Example:

A Wi-Fi router connects your home network to the Internet.

Real-life Analogy:

- A traffic police officer directing vehicles to different roads.



Higher Layer Device (Layer 4–7)

7) Gateway

A gateway is a device that connects networks using different communication protocols.

Function:

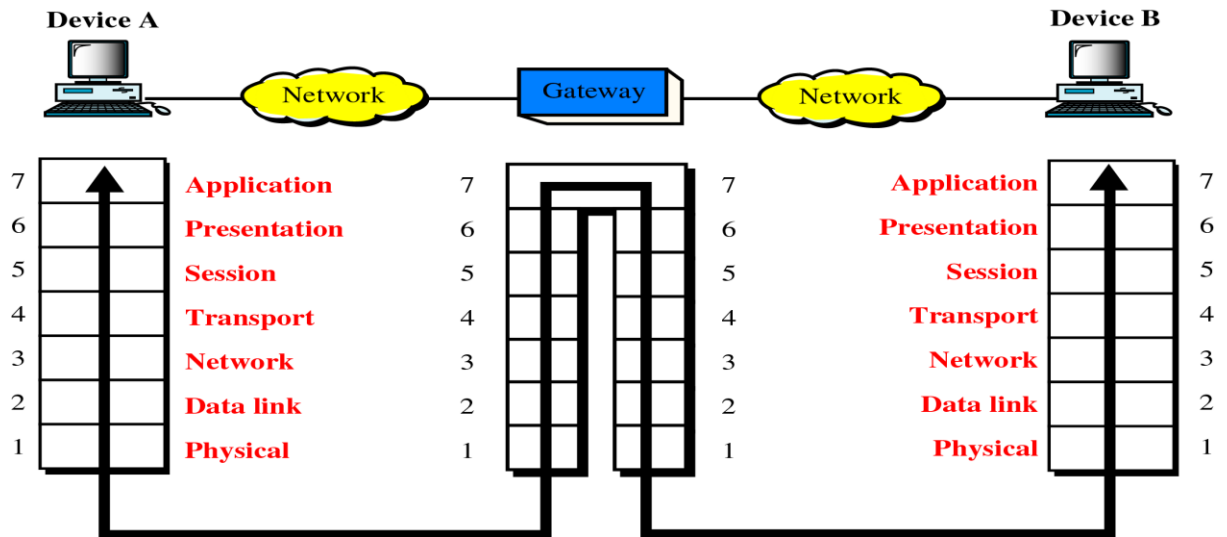
Translates data between incompatible networks.

How it Works:

Converts data formats and protocols for communication.

Key Points:

- Can operate at multiple OSI layers
- Acts as protocol converter



Example:

An email gateway allows communication between internal mail systems and the Internet.

Real-life Analogy:

An international airport connecting flights between different countries.

COMPARISON: NETWORKING DEVICES

Device	OSI Layer	Main Function / Key Difference
Hub	Layer 1 (Physical)	Broadcasts data to all devices ; no intelligence
Repeater	Layer 1 (Physical)	Regenerates and amplifies weak signals
Modem	Layer 1 (Physical)	Converts digital ↔ analog signals for Internet
Bridge	Layer 2 (Data Link)	Filters traffic using MAC addresses
Switch	Layer 2 (Data Link)	Forwards data only to the destination device
Router	Layer 3 (Network)	Routes data between networks using IP addresses
Gateway	Layer 4–7 (Multiple)	Converts data between different protocols